S06: Is the control advantage of high-internal/low-external participants specific to race?

Question. Among people high in internal motivation to respond without prejudice, do those low in external motivation (HI/LE) show better response control than those high in it (HI/HE) specifically on the race-primed gun/tool task (Weapons) rather than on a race-neutral flanker task? Tested as the Group × Task interaction on the process-dissociation (PD) control estimate.

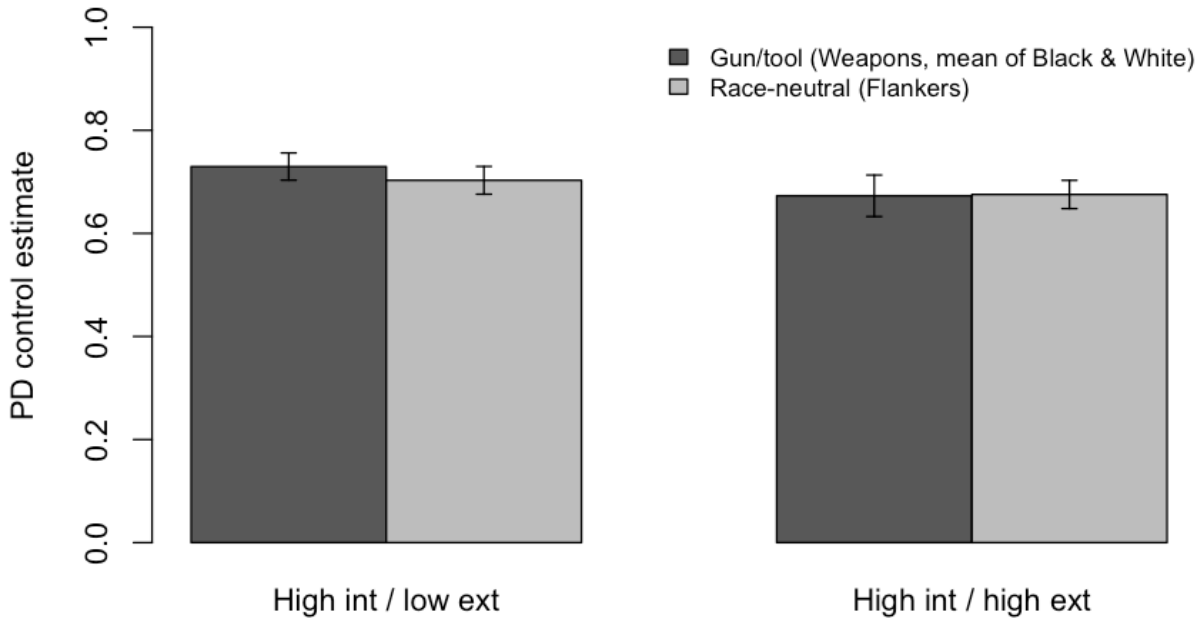
Data and sample. Replication's derived PD file (`d11.csv`): three rows per participant (Weapons-Black, Weapons-White, Flankers). N = 75 (46 HI/LE, 29 HI/HE), 225 rows, no missing values. **No exclusions** (as pre-specified on the card); 0 participants removed.

Pre-specified primary test. `aov(PD ~ Group * Task + Error(ID/Task))` in R, both Weapons rows kept; Group × Task F from the ID:Task stratum; partial η^2 = SS_int / (SS_int + SS_res). Criterion for "holds": $p < .05$ with the group advantage (HI/LE − HI/HE) larger on Weapons than on Flankers.

Descriptives (mean PD, SD).

Group	Weapons-Black	Weapons-White	Weapons (mean)	Flankers
HI/LE (n = 46)	.743 (.180)	.717 (.190)	.730	.703 (.182)
HI/HE (n = 29)	.702 (.214)	.644 (.232)	.673	.675 (.148)
Difference	.040	.073	.057	.028

Result. Group × Task: **$F(1, 73) = 0.43$, $p = .513$, partial $\eta^2 = .006$ (90% CI [.000, .065]; 95% CI [.000, .083])**. The direction is as predicted (group difference .057 on Weapons vs .028 on Flankers; interaction contrast .029, 95% CI [−.059, .118] on the PD scale), but it is small and far from significant. Neither simple group effect is significant (Weapons: Welch $t(51.3) = 1.18$, $p = .244$; Flankers: $t(68.4) = 0.72$, $p = .476$), nor is the Group main effect ($F(1, 73) = 1.45$, $p = .232$).



Checks. The 2 × 2 mixed ANOVA on the per-participant Weapons mean gives the identical interaction ($F(1, 73) = 0.43$, $p = .513$), as does the pingouin cross-check in Python; a Wilcoxon test on the Weapons–Flankers difference score gives $p = .476$. On the Weapons task, the HI/LE advantage was, if anything, larger on White-face trials (.073) than on Black-face trials (.040), which does not fit a race-bias-specific account.

Limitations. Unequal and modest groups (46/29); with this N the 95% CI on partial η^2 still includes effects up to about .08, so a small-to-moderate interaction cannot be ruled out.

Conclusion. The claim does not hold on this data: the HI/LE control advantage is not detectably larger on the race-primed gun/tool task than on the race-neutral task ($F(1, 73) = 0.43$, $p = .51$, partial $\eta^2 = .006$).

Files: `analysis.py` (runs `analysis.R`), `analysis.out`, `log.md`.